29 Multi-Scale Trajectory Optimisation for Ground-Based SLAM in Archaeological Contexts: Mitigating Noise and Maximising Data Yield Across Confined and Open Excavations

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The deployment of multi-sensor ground-based Simultaneous Localization and Mapping (SLAM) systems in archaeological contexts introduces critical challenges regarding spatial trajectory optimization. Data integrity is strictly dependent on the operator's kinematic path, which must dynamically adapt to highly divergent spatial constraints. This study presents a structured acquisition pipeline designed to formalize path planning across two distinct dimensional paradigms: confined target areas (tombs, hypogeal environments) and expansive open-area excavations. The proposed methodology models the optimal routing protocol to prevent SLAM algorithm divergence and reduce cumulative drift. For micro-scale environments characterized by tight geometric constraints, the protocol mandates specific loop-closure frequencies and controlled rotational velocities to manage close-range sensor saturation and occlusion phenomena. Conversely, in macro-scale scenarios such as open trenches, the pipeline dictates overlapping transects integrated with strategic geometric anchors, maintaining global registration accuracy over extended temporal and spatial bounds. By formalizing these spatial routing parameters, the developed multi-sensor SLAM protocol systematically minimizes algorithmic noise while maximizing volumetric data capture. Ultimately, this methodology shifts ground-based SLAM from an improvised scanning procedure to a rigorous, replicable acquisition framework, ensuring high-fidelity 3D documentation across heterogeneous archaeological topologies.